

SVETLOGRAD, Russian Federation

WMO: 349540

Lat: 45.350N

Lon: 42.850E

Elev: 462

StdP: 14.45

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.2	(c) 11.7	(d) -1.0	(e) 5.3	(f) 6.0	(g) 5.6	(h) 7.5	(i) 13.2	(j) 13.5	(k) 35.9	(l) 11.3	(m) 34.6	(n) 2.1	(o) 130	(p) 0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	95.2	68.0	92.1	67.4	88.8	66.8	71.8	86.4	70.4	84.6	69.1	82.8	4.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	103.0	76.7	66.1	97.9	75.0	64.7	93.1	74.2	35.8	86.5	34.6	84.1	33.5	83.2	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.4	7.7	-0.9	99.2	8.9	3.2	-7.3	101.5	-12.5	103.4	-17.4	105.2	-23.9	107.6
			-1.5	75.1	8.7	1.6	-7.8	76.3	-12.9	77.2	-17.8	78.0	-24.1	79.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	30.3	32.7	40.8	52.7	63.4	70.5	76.4	74.9	64.7	53.4	41.0	32.6		
	DBStd	18.04	8.77	9.47	7.63	7.71	7.01	5.76	4.96	5.16	6.80	8.80	7.95	9.30		
	HDD50	2343	610	486	299	57	3	0	0	0	2	67	280	539		
	HDD65	5416	1075	905	751	371	115	18	0	1	87	371	719	1003		
	CDD50	3405	0	1	13	139	420	614	819	773	444	171	11	0		
	CDD65	1005	0	0	0	4	67	183	354	309	78	10	0	0		
	CDH74	10330	0	0	1	60	701	1714	3902	3201	645	106	0	0		
	CDH80	4274	0	0	0	13	176	620	1870	1422	151	22	0	0		
(14) Wind	WSAvg	2.8	3.0	3.3	3.7	3.2	2.8	2.7	2.3	2.2	2.4	2.2	3.0	3.1		
(15) Precipitation	PrecAvg	19.0	1.0	0.9	1.2	1.7	2.2	2.8	2.0	1.9	1.3	1.7	1.3	1.2		
	PrecMax	27.3	2.1	2.0	3.3	4.8	4.1	5.4	8.7	5.4	2.8	4.0	2.7	3.7		
	PrecMin	11.9	0.0	0.1	0.1	0.4	0.9	0.9	0.2	0.0	0.2	0.1	0.3	0.1		
	PrecStd	4.1	0.5	0.6	0.8	1.2	0.9	1.5	1.9	1.5	0.8	1.1	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	58.7	70.4	80.7	87.8	94.9	99.7	98.1	88.1	82.6	65.2	54.1		
		MCWB	43.7	47.0	52.7	59.5	64.3	67.1	68.6	69.0	66.4	61.8	54.2	47.3		
	2%	DB	45.8	52.5	62.2	74.4	84.4	90.4	96.6	94.3	84.3	76.4	58.7	49.6		
		MCWB	41.4	44.5	49.9	57.7	63.0	67.0	68.2	68.2	64.8	60.8	52.0	44.6		
	5%	DB	42.8	47.2	57.2	70.8	81.6	86.7	93.2	91.2	80.9	71.2	54.8	46.6		
		MCWB	39.2	41.4	47.1	56.0	62.6	65.9	67.3	67.5	63.9	59.3	50.0	42.8		
	10%	DB	40.2	43.3	52.8	66.8	77.9	83.2	89.7	87.9	77.6	66.0	51.7	44.1		
		MCWB	37.6	39.4	45.1	54.0	61.5	65.3	66.9	67.4	63.0	57.0	48.0	41.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	48.6	53.4	61.6	68.2	72.9	72.9	74.0	69.7	64.5	55.7	48.3		
		MCDB	49.4	57.0	67.2	75.5	80.9	86.8	89.5	87.2	81.5	76.6	61.2	53.6		
	2%	WB	41.9	44.8	50.7	59.5	66.2	70.1	71.1	72.1	67.5	62.2	53.0	45.4		
		MCDB	45.1	51.3	60.5	72.1	78.8	84.1	86.4	86.3	78.4	72.6	57.7	49.0		
	5%	WB	39.6	41.9	48.2	57.4	64.4	68.3	69.9	70.8	66.1	60.2	50.4	43.2		
		MCDB	42.3	46.2	55.2	68.0	76.6	81.2	85.6	84.5	76.6	69.0	53.9	46.2		
	10%	WB	37.7	39.6	46.0	55.0	62.8	66.8	68.7	69.4	64.6	58.3	48.2	41.3		
		MCDB	40.0	43.1	52.3	64.0	74.1	79.3	84.0	82.0	75.0	65.3	51.6	43.8		
(35) Mean Daily Temperature Range	MDBR		10.0	12.2	14.6	18.8	20.1	19.7	21.9	21.6	20.2	16.7	11.9	10.6		
	5% DB	MCDBR	14.6	19.3	24.5	25.3	26.7	26.4	27.7	27.2	25.3	25.1	19.6	14.5		
		MCWBR	10.6	11.8	13.4	11.4	10.1	9.2	7.6	8.2	9.8	11.7	12.3	10.1		
	5% WB	MCDBR	13.0	17.8	21.1	22.2	22.0	22.0	22.7	21.7	20.7	22.1	17.5	13.2		
		MCWBR	9.7	11.5	12.3	10.5	9.5	9.0	7.7	8.1	9.1	11.4	11.4	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.375	0.415	0.461	0.428	0.434	0.458	0.459	0.429	0.390	0.355	0.345		
	taud		2.252	2.172	2.118	2.029	2.214	2.241	2.185	2.172	2.217	2.288	2.320	2.259		
	Ebn at Noon		234	248	255	253	267	266	257	252	248	241	229	219		
	Edh at Noon		27	36	43	51	44	43	45	44	39	31	25	24		
(44) All-Sky Solar Radiation	RadAvg		361	635	939	1469	1840	2046	2040	1838	1340	800	424	292		
	RadStd		86	109	97	142	184	116	116	89	130	104	58	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	N/A

Nomenclature: See separate page